Norm-Attaining Operators of Every Finite Rank on Martín's Space

A verification and solution report

June 22, 2026

Abstract

Miguel Martín constructed a smooth renorming $\mathfrak{X}$ of c_0 whose set of norm-attaining functionals contains no non-trivial cone, and asked whether $\mathrm{NA}(\mathfrak{X}, \ell_2^2)$ nevertheless contains an operator of rank two. We first audit a proposed fixed-kernel nowhere-density argument. Its conclusion is correct, but the written proof uses the false general assertion that smoothness of a Banach space is equivalent to strict convexity of its dual. A local quotient-smoothness argument repairs the proof.

We then give a positive answer to Martín's question. In fact, a stronger statement follows directly from the architecture of his renorming: for every integer $n \geq 1$ and every real Banach space Y with $\dim Y \geq n$, there is a norm-attaining operator from $\mathfrak{X}$ to Y of rank n. The key observation is a general lifting lemma for a graph-sum norm

$$p(x) = \|x\|_0 + \lambda \|Rx\| .$$

A finite-rank norm-attaining operator for the old norm can be modified by one rank-one term in the direction of its norming value. The triangle inequality then gives the upper bound for the new operator norm, while equality holds at the same vector. Martín's injectivity and range-separation properties ensure that the rank is preserved.

1 The question and the literature check

Throughout, all Banach spaces and operators are real. For Banach spaces X, Y, $\mathrm{NA}(X, Y)$ denotes the set of norm-attaining bounded operators from X to Y.

Martín's paper constructs a particular renorming $\mathfrak{X}$ of c_0 and ends with the following question:

Does $\mathrm{NA}(\mathfrak{X}, \ell_2^2)$ *contain rank-two operators?*

This is Open Problem 7 in [1]. A targeted search through June 22, 2026 found no published proof or counterexample. In particular, Martín's May 2025 BIRS handout still states the question as open [3]. Thus the argument below is presented as a new solution and has not been peer reviewed.

Our main result is stronger than the requested rank-two assertion.

Theorem 1.1 (Main theorem)**.** *Let $\mathfrak{X}$ be the Banach space produced in the proof of [1, Theorem 1]. Let $n \in \mathbb{N}$, and let Y be a real Banach space with $\dim Y \geq n$. Then there exists $T \in \mathrm{NA}(\mathfrak{X}, Y)$ with*

$$\operatorname{rank} T = n.$$

In particular, $\mathrm{NA}(\mathfrak{X}, \ell_2^2)$ contains a rank-two operator.

2 The structural ingredients in Martín's construction

We isolate only the properties actually used below. In Steps 1–7 of the proof of [1, Theorem 1], the final space $\mathfrak{X}$ is built as follows.

The underlying vector space is c_0. First it is given an equivalent smooth norm $\|\cdot\|_0$; write

$$X_0 = (c_0, \|\cdot\|_0).$$

The construction has

$$\mathrm{NA}(X_0, \mathbb{R}) = \mathrm{NA}(c_0, \mathbb{R}) = c_{00}. \tag{1}$$

Here the dual spaces are identified algebraically with ℓ_1. Recall that an element of $\ell_1 = c_0^*$ attains its usual norm on c_0 exactly when it has finite support, which proves the second equality in (1).

There are also a Banach space W, a bounded operator $R : X_0 \to W$, and a linear subspace $\mathscr{Y} \subset X_0^*$ such that

$$R \text{ is injective,} \tag{2}$$

$$R^* : W^* \longrightarrow X_0^* \text{ is injective,} \tag{3}$$

$$R^*(W^*) \subseteq \mathscr{Y}, \tag{4}$$

$$\mathscr{Y} \cap \mathrm{NA}(X_0, \mathbb{R}) = \{0\}. \tag{5}$$

Specifically, (4) and (3) are equations (3) and (4) in Step 6 of Martín's proof. The final norm is

$$p(x) = \|x\|_0 + \|Rx\|_W, \qquad x \in c_0, \tag{6}$$

and $\mathfrak{X} = (c_0, p)$. The scaled variants in [1, Remark 2] have $p_\lambda(x) = \|x\|_0 + \lambda \|Rx\|$, where $\lambda > 0$; the same argument works for all of them.

3 Two elementary lemmas

We first record the finite-rank norm-attainment criterion used in [2, Corollary 2.6]. A short proof is included.

Lemma 3.1 (Finite-rank criterion). *Let X, Y be Banach spaces and let $S : X \to Y$ have finite rank. If*

$$[\ker S]^\perp = \operatorname{ran} S^* \subseteq \mathrm{NA}(X, \mathbb{R}),$$

then $S \in \mathrm{NA}(X, Y)$.

Proof. Put $E = \operatorname{ran} S$, and regard S as a surjection from X onto the finite-dimensional space E. The unit sphere of E^* is compact, so there is $e^* \in S_{E^*}$ such that

$$\|S^* e^*\| = \|S^*\| = \|S\|.$$

The functional $S^* e^*$ belongs to $[\ker S]^\perp$, hence attains its norm at some $x \in S_X$. After changing the sign of x if needed,

$$\|S\| = (S^* e^*)(x) = e^*(Sx) \leq \|Sx\| \leq \|S\|.$$

Thus $\|Sx\| = \|S\|$. ☐

The next lemma is the mechanism that passes from the old norm to the graph-sum norm.

Lemma 3.2 (Graph-sum lifting)**.** *Let $(X, \|\cdot\|_0)$ and W be Banach spaces, let $R : X \to W$ be bounded, and fix $\lambda > 0$. Define*

$$p_\lambda(z) = \|z\|_0 + \lambda \|Rz\| .$$

Suppose that $S : X \to Y$ has finite rank, $\|S\| = 1$ for the old norm, and attains its norm at $x \in S_{(X,\|\cdot\|_0)}$. Put $y = Sx$, so $\|y\| = 1$. Assume that $Rx \neq 0$, and choose $w^* \in S_{W^*}$ with

$$w^*(Rx) = \|Rx\| .$$

If

$$R^*w^* \notin \operatorname{ran} S^*, \tag{7}$$

then the operator

$$Tz = Sz + \lambda (R^*w^*)(z)y, \qquad z \in X, \tag{8}$$

viewed from (X, p_λ) to Y, satisfies

$$T \in \mathrm{NA}((X, p_\lambda), Y), \qquad \|T\| = 1, \qquad \operatorname{rank} T = \operatorname{rank} S.$$

It attains its norm at $x/p_\lambda(x)$.

Proof. For every $z \in X$, the triangle inequality gives

$$\begin{aligned} \|Tz\| &\leq \|Sz\| + \lambda |w^*(Rz)| \|y\| \\ &\leq \|z\|_0 + \lambda \|Rz\| = p_\lambda(z). \end{aligned}$$

Hence $\|T\| \leq 1$. At the norming vector x,

$$Tx = y + \lambda \|Rx\| \, y = \big(1 + \lambda \|Rx\|\big)y,$$

whereas

$$p_\lambda(x) = 1 + \lambda \|Rx\| .$$

Consequently $\|T\| = 1$, and T attains its norm at $x/p_\lambda(x)$.

It remains to check the rank. Let $E = \operatorname{ran} S$; then $y \in E$ and $\operatorname{ran} T \subseteq E$. Regard S, T as operators into E. Since $S : X \to E$ is onto, $S^* : E^* \to X^*$ is injective and $\operatorname{ran} S^* = [\ker S]^\perp$. If $T^*e^* = 0$ for some $e^* \in E^*$, then

$$S^*e^* = -\lambda e^*(y) R^*w^*.$$

If $e^*(y) = 0$, injectivity of S^* gives $e^* = 0$. If $e^*(y) \neq 0$, the displayed identity contradicts (7). Thus $T^* : E^* \to X^*$ is injective, so $T : X \to E$ is onto. Therefore $\operatorname{rank} T = \dim E = \operatorname{rank} S$. □

4 Proof of the main theorem

Proof of Theorem 1.1. Fix $n \in \mathbb{N}$ and a Banach space Y with $\dim Y \geq n$. Choose an n-dimensional subspace $E \subseteq Y$ and an n-dimensional subspace

$$U \subset c_{00} = \mathrm{NA}(X_0, \mathbb{R}).$$

Let $M = U_\perp = \bigcap_{f \in U} \ker f$. Then X_0/M is n-dimensional and

$$M^\perp = U.$$

Choose a linear isomorphism $A : X_0/M \to E$, let $q : X_0 \to X_0/M$ be the quotient map, and define

$$S_0 = \iota_E A q : X_0 \longrightarrow Y,$$

where $\iota_E : E \hookrightarrow Y$ is the inclusion. Then $\operatorname{rank} S_0 = n$, $\ker S_0 = M$, and

$$\operatorname{ran} S_0^* = M^\perp = U \subseteq \mathrm{NA}(X_0, \mathbb{R}).$$

By Lemma 3.1, S_0 attains its norm. Replace it by $S = S_0/\|S_0\|$, choose $x \in S_{X_0}$ with $\|Sx\| = 1$, and put $y = Sx$.

Since R is injective and $x \neq 0$, one has $Rx \neq 0$. Choose $w^* \in S_{W^*}$ satisfying $w^*(Rx) = \|Rx\|$, and set

$$r = R^* w^*.$$

By (4), $r \in \mathscr{Y}$. By injectivity of R^*, $r \neq 0$. If $r \in \operatorname{ran} S^* = U$, then

$$r \in \mathscr{Y} \cap U \subseteq \mathscr{Y} \cap \mathrm{NA}(X_0, \mathbb{R}) = \{0\},$$

a contradiction. Hence $r \notin \operatorname{ran} S^*$.

Lemma 3.2, with $\lambda = 1$, now shows that

$$Tz = Sz + r(z)y.$$

more simply written as

$$T = S + y \otimes r, \qquad (y \otimes r)(z) = r(z)y,$$

is a norm-one norm-attaining operator from $(c_0, p) = \mathfrak{X}$ to Y, and $\operatorname{rank} T = \operatorname{rank} S = n$. This proves the theorem. □

Remark 4.1 (The rank-two formula)**.** *For $n = 2$ and $Y = \ell_2^2$, rotate the norm-one base operator so that*

$$S = (f_0, h_0), \qquad Sx = (1, 0).$$

Then the resulting operator has the especially transparent form

$$Tz = \big(f_0(z) + (R^* w^*)(z),\ h_0(z)\big).$$

Indeed,

$$\|Tz\|_2 \leq \|(f_0(z), h_0(z))\|_2 + |w^*(Rz)| \leq \|z\|_0 + \|Rz\| = p(z),$$

with equality at $x/p(x)$. The two coordinate functionals are independent because $R^* w^* \notin \operatorname{span}\{f_0, h_0\}$.

Corollary 4.2. *The conclusion of Theorem 1.1 also holds for every scaled norm*

$$p_\lambda(x) = \|x\|_0 + \lambda \|Rx\|, \qquad \lambda > 0,$$

including the arbitrarily small equivalent perturbations described in [1, Remark 2].

Proof. Use Lemma 3.2 with the same S, x, y, w^* and the specified λ. □

5 Audit and repair of the fixed-kernel partial result

The proposed partial result asserts that for every fixed two-codimensional kernel, norm attainment is nowhere dense in the corresponding four-dimensional operator fibre. The conclusion is correct, but two points in the supplied write-up need correction.

First, the finiteness statement for $q(B_{\mathfrak{X}}) \cap S_{\mathfrak{X}/M}$ is supplied by Proposition 6(b) of [1], not by "Proposition 2.4" of that paper. Second, the sentence

"$\mathfrak{X}$ is smooth; equivalently, $\mathfrak{X}^*$ is strictly convex"

is false for general Banach spaces. In infinite dimensions, smoothness of a space does not in general imply strict convexity of its dual. Consequently, the global claim that every two-dimensional quotient $\mathfrak{X}/M$ is smooth does not follow from the stated reasoning.

Only smoothness at a finite set of quotient points is needed, and that property follows directly from smoothness of $\mathfrak{X}$.

Lemma 5.1 (Local smoothness passes to a represented quotient point)**.** *Let $q : X \to E = X/M$ be a quotient map. Suppose that $x \in S_X$, that $e = qx \in S_E$, and that X is smooth at x. Then E is smooth at e.*

Proof. Let $\phi_1, \phi_2 \in S_{E^*}$ satisfy $\phi_1(e) = \phi_2(e) = 1$. The adjoint $q^* : E^* \to X^*$ is an isometry, and

$$(q^*\phi_i)(x) = \phi_i(qx) = 1 \qquad (i = 1, 2).$$

Thus both $q^*\phi_1$ and $q^*\phi_2$ are norming functionals for x. Smoothness at x gives $q^*\phi_1 = q^*\phi_2$. Since q^* is injective, $\phi_1 = \phi_2$. □

We now give a corrected proof of the partial theorem. The finite-dimensional perturbation lemma needs smoothness only at the prescribed vector.

Lemma 5.2. *Let E be a real two-dimensional Banach space, let $H = \ell_2^2$, and fix $e \in S_E$. Assume that E is smooth at e. Define*

$$F_e = \{A \in \mathcal{L}(E, H) : \|Ae\| = \|A\|\}.$$

Then F_e is closed and nowhere dense in $\mathcal{L}(E, H)$.

Proof. Closedness is immediate. We show that F_e has empty interior.

Let $A \in F_e$ and $\delta > 0$. If $A = 0$, choose a nonzero $B : E \to H$ with $Be = 0$ and $\|B\| < \delta$; then $B \notin F_e$.

Assume $A \neq 0$, put $y = Ae$, and let $j \in S_{E^*}$ be the unique norming functional for e. Choose $d \in \ker j \setminus \{0\}$. The norm of E is Gâteaux differentiable at e, so

$$\left.\frac{d}{dt}\right|_{t=0} \|e + td\| = j(d) = 0.$$

Set $u = y/\|y\|$. Since $\|A(e + td)\| \le \|y\|\,\|e + td\|$ with equality at $t = 0$, differentiation from both sides gives

$$\langle u, Ad \rangle = 0.$$

Choose $B \in \mathcal{L}(E, H)$ with $Be = 0$ and $Bd = y$, and put $A_\alpha = A + \alpha B$. Then $A_\alpha e = y$, while

$$A_\alpha(e + td) = y + tAd + \alpha t y.$$

The derivative at zero of the numerator in

$$\frac{\|A_\alpha(e+td)\|}{\|e+td\|}$$

is

$$\langle u, Ad+\alpha y\rangle = \alpha\,\|y\| > 0,$$

and the derivative of the denominator is zero. Hence, for all sufficiently small $t>0$,

$$\frac{\|A_\alpha(e+td)\|}{\|e+td\|} > \|y\| = \|A_\alpha e\|\,.$$

Thus $A_\alpha \notin F_e$. Taking $\alpha>0$ so small that $\|\alpha B\|<\delta$ proves that F_e has empty interior. □

Theorem 5.3 (Corrected fixed-kernel theorem)**.** *Let M be a closed subspace of $\mathfrak{X}$ with* $\operatorname{codim} M = 2$*, let $H=\ell_2^2$, and set*

$$\mathcal{L}_M = \{T\in\mathcal{L}(\mathfrak{X},H) : M\subseteq \ker T\}.$$

Then $\mathrm{NA}(\mathfrak{X},H)\cap\mathcal{L}_M$ is nowhere dense in $\mathcal{L}_M$.

Proof. Let $q:\mathfrak{X}\to E=\mathfrak{X}/M$ be the quotient map. The map $A\mapsto Aq$ identifies $\mathcal{L}(E,H)$ isometrically with $\mathcal{L}_M$. Set

$$C_M = q(B_{\mathfrak{X}})\cap S_E.$$

By [1, Proposition 6(b)], C_M contains at most two antipodal pairs.

For each $e\in C_M$, choose $x\in B_{\mathfrak{X}}$ with $qx=e$. Since $\|e\|=1$, necessarily $\|x\|=1$. The space $\mathfrak{X}$ is smooth, so Lemma 5.1 shows that E is smooth at e. Lemma 5.2 therefore applies to every $e\in C_M$.

We claim that

$$\mathrm{NA}(\mathfrak{X},H)\cap\mathcal{L}_M \subseteq \{0\}\cup\bigcup_{e\in C_M} F_e. \tag{9}$$

Indeed, let $T=Aq\neq 0$ attain its norm at $x\in S_{\mathfrak{X}}$. Since $\|Aq\|=\|A\|$,

$$\|A(qx)\| = \|A\|\,.$$

Writing $e=qx$, one cannot have $\|e\|<1$, because then

$$\|A(e/\|e\|)\| = \frac{\|Ae\|}{\|e\|} > \|A\|\,.$$

Thus $e\in q(B_{\mathfrak{X}})\cap S_E = C_M$, and $A\in F_e$. This proves (9). The right-hand side is a finite union of closed nowhere dense subsets of the four-dimensional space $\mathcal{L}(E,H)$, so the left-hand side is nowhere dense. □

Corollary 5.4. *If $T\in\mathrm{NA}(\mathfrak{X},\ell_2^2)$ has rank two and kernel M, then for every $\varepsilon>0$ there is a rank-two operator $S:\mathfrak{X}\to\ell_2^2$ such that*

$$\ker S = M,\qquad \|S-T\|<\varepsilon,\qquad S\notin\mathrm{NA}(\mathfrak{X},\ell_2^2).$$

Proof. Write $T=Aq$, where $q:\mathfrak{X}\to\mathfrak{X}/M$ is the quotient map. Since T has rank two, $A:\mathfrak{X}/M\to\ell_2^2$ is invertible. Invertible operators form an open subset of $\mathcal{L}(\mathfrak{X}/M,\ell_2^2)$. Theorem 5.3 therefore supplies, arbitrarily close to A, an invertible B such that Bq is not norm attaining. Put $S=Bq$. Then $\ker S=M$, $\operatorname{rank} S=2$, and $\|S-T\|=\|B-A\|$. □

6 How the two conclusions fit together

Theorem 1.1 and Theorem 5.3 are fully compatible. The main theorem proves existence of many finite-rank norm-attaining operators, including rank-two ones. The fixed-kernel theorem says that after one fixes the kernel of any such operator, norm attainment is still a nowhere dense condition inside that particular four-dimensional fibre. Non-emptiness and nowhere density are not contradictory.

The argument settles only the existence question in Open Problem 7. It does not by itself settle the older density problem asking whether $\mathrm{NA}(X, \ell_2^2)$ is dense in $\mathcal{L}(X, \ell_2^2)$ for every Banach space X, nor whether every finite-rank operator can be approximated by norm-attaining operators.

Conclusion

The candidate fixed-kernel result survives audit after replacing an incorrect duality assertion by a local quotient-smoothness lemma. More importantly, the graph-sum structure of Martín's norm yields a direct positive solution of Open Problem 7, and in fact produces norm-attaining operators of every prescribed finite rank into every sufficiently large range space.

References

[1] M. Martín, *A Banach space whose set of norm-attaining functionals is algebraically trivial*, J. Funct. Anal. **288** (2025), 110815. Preprint: arXiv:2406.07273.

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